Spinning and Boosted Trumpet Initial Data: Formulation and Validation Status

TwoPuncturesC and AthenaK integration

(Dated: 18 October 2026 — interim report)

Abstract

We implement quasi-isotropic (QI), $R_0 = M$ trumpet and Liu wormhole seeds with portable AthenaK import. First Cartesian focus compatibility reduces the finest moderate QI bulk momentum RMS by 21.9 to 7.77×10^{-9}. **The strict bulk-convergence gate remains failed.** One fixed three-grid formulation meets absolute physical constraint bounds; a calibrated sixth-order observer confirms the residual rise. Stronger second compatibility worsens the error and is rejected. Finest-grid AthenaK horizons pass angular, quadrature and interior-enclosure checks. Matched-count Bowen–York Hamiltonian RMS is near 10^{-10}, at the observer calibration floor; failed native stopping gates are retained. Isolated Liu $\chi = 0.99$ controls pass. Moderate trumpet and extreme binaries remain unqualified.

I. CONSTRUCTION AND ITS LIMITS

The reference is the $R_0 = M$ stationary Kerr trumpet of Dennison, Baumgarte and Montero, Eq. 12 of arXiv:1409.1887v2. The reused AthenaK reference is pinned to `project/kerr-trumpet-spin09`, commit `81c7d8030c49489577c54b977b8b7f7b42d9a96e`. This is distinct from both Ian Ruchlin's quasi-isotropic wormhole HiSpID free data and a stationary standard $1 + \log$ trumpet. The existing quasi-isotropic family remains the default; residual-Z4c evolution background subtraction is not added to the elliptic equations.

We use signature $(-+++)$ and $K_{ij} = -\mathcal{L}_n \gamma_{ij}/2$. For rest mass m, $a = |\boldsymbol{S}|/m$, and $R = r + m$, one scalar choice is $\psi_0 = (\Sigma/r^2)^{1/4}$ with $\Sigma = R^2 + a^2 \cos^2\theta$; the other is $(\det\gamma)^{1/12}$. Both have direction-dependent $r^{-1/2}$ puncture behavior. Full spacetime Lorentz pullbacks supply boosted spatial tensors and their time derivatives. A rest-chart connection contraction gives the extrinsic curvature stably. Spacelikeness and positive spatial metrics are checked without clipping. Nonspinning boosted slicing has an analytic global spacelikeness argument; general spinning cases retain sampled-domain checks.

The four elliptic unknowns are a scalar correction u and vector b^i. Outside the explicitly modified interior, the CTT equations are

$$\widetilde{\Delta}\psi - \tfrac{1}{8}\widetilde{R}\psi - \tfrac{1}{12}K^2\psi^5 + \tfrac{1}{8}\widetilde{A}_{ij}\widetilde{A}^{ij}\psi^{-7} = 0,$$
$$\widetilde{D}_j\widetilde{A}^{ij} - \tfrac{2}{3}\psi^6\widetilde{D}^i K = 0, \qquad \widetilde{A}^{ij} = \widetilde{A}_0^{ij} + (\widetilde{L}b)^{ij}.$$

The covariant CTT tensor is $\widetilde{A}_{ij} = \psi^2(K_{ij} - \gamma_{ij}K/3)$; AthenaK's Z4c conformal tensor

instead uses ψ^{-4} on the same physical trace-free tensor. The import explicitly converts these conventions.

Unmodified trumpet corrections cannot simply be assumed analytic at the puncture: the isolated Schwarzschild scalar indicial exponent includes $p_0 = \sqrt{3} - 1/2$. The implemented binary examples use the declared interior plateau, where the correction equations become $\Delta u = 0$ and $\Delta b + \nabla \operatorname{div} b/3 = 0$. Bounded corrections then have removable singularities. This modifies the interior operator; vacuum accuracy is tested independently outside it, and measured horizons must enclose the modified regions. It is not evidence of exponential convergence across the transition.

II. ACCEPTANCE AND PRODUCTION PATH

The production stages are host extended-precision geometry/cache construction, CUDA GMRES for the solve, CPU sampling, and OpenMP AthenaK horizon integration. The opt-in no-swirl axisymmetric Krylov projection is restricted to nonspinning axial configurations; full unprojected nonlinear residuals still govern acceptance. Default QI and Bowen–York paths are preserved. No backend benchmark matrix is part of this investigation.

Obligation	Decisive criterion	Status
Seeds and boosts	Independent fields/derivatives, vacuum checks, charges, positive slicing; generic orientation included	Pass within tested domain
Operators	Manufactured curved operators and centered JVP checks	Pass
Binary constraints	Near and bulk RMS $< 10^{-6}$, maximum $< 10^{-4}$; three-grid decrease, positive ψ, stable charges	Not passed
Import	Checkpoint-bound fields and gradients within 10^{-12}	Passed cases
Horizons	Expansion RMS $< 10^{-7}$; angular/quadrature stability, mass/spin changes $< 10^{-4}$; enclosure	See below
Compatibility	Historical QI fixture; original batch-seed API remains bitwise identical	See update

The physical observers are independent of the weighted collocation residual. Modified-region errors are recorded separately. A successful solve or import does not promote diagnostic data

to physical acceptance. Checkpoints preserve family, basis/map and producer identity; v1 retains its QI meaning and v2 adds explicit family metadata.

III. LATEST REGULARITY INVESTIGATION: 18 OCTOBER

The unmodified Schwarzschild scalar sector admits $u \sim r^{\sqrt{3}-1/2}$, so factoring out the seed singularity does not imply an analytic correction. The flat interior plateau removes that local scalar potential, but not possible regularity limitations at its nonanalytic compact transition or at infinity. This isolated scalar exponent is not the full coupled spinning/boosted indicial spectrum.

Offline analysis of retained $160 \times 320 \times 16$ factored $m = 0$ unknowns finds slow meridional coefficient tails in both seed families. Degree-eight polynomial windows localize the wider-interior trumpet's radial vector tail mainly toward compactified infinity: suppressing that endpoint leaves 0.00406 of the degree 80–143 band norm, whereas suppressing the inner radial boundary leaves 1.003. The narrow QI control has the opposite localization. The inner boundary includes both foci and the axis segment between them. These are representation diagnostics of distinct interiors, not a controlled slicing-only comparison or an identification of logarithmic powers. No filtered field is used as a solution.

The QI control retains masses 0.6/0.4, separation 6, and the original generic spin and velocity vectors ($|\chi| = 0.3905/0.3742$). It uses the exact seed scalar choice, no far attenuation, the same mapped modal basis, and the original modal GMRES on CUDA. The narrow interior window spans fractions 0.15–0.5 of the isolated QI throat radius. Its 160-grid solve converges internally in 8 Newton and 46 Krylov iterations, but bulk momentum RMS 3.809×10^{-6} fails the 10^{-6} bound. A separate candidate widens the outer fraction to 0.85, retaining the 0.15 plateau. At unchanged exterior points this reduces bulk momentum RMS to 1.045×10^{-7}. Its fixed sequence is:

Grid	Near H	Near M	Bulk H	Bulk M	Krylov
$160 \times 320 \times 16$	3.13×10^{-7}	9.14×10^{-8}	6.30×10^{-9}	1.05×10^{-7}	35
$192 \times 384 \times 16$	1.44×10^{-7}	4.29×10^{-8}	8.62×10^{-9}	1.09×10^{-7}	34
$224 \times 448 \times 16$	4.88×10^{-8}	2.03×10^{-8}	5.32×10^{-8}	1.70×10^{-7}	251

All three use the same native image and physical free data, and all pass the absolute

near/bulk RMS and maximum bounds. The 48 near-hole and 18 bulk points are fixed, including all finite-difference stencil points in the unmodified region. The nonlinear weighted maximum is below 10^{-12} on every grid; ψ and spatial metrics are positive at the declared observers. Extrapolated charge changes, scaled by $1+|Q_{\mathrm{finer}}|$, are 3.83×10^{-10} and 2.35×10^{-9}, below 10^{-3}. **Bulk errors do not decrease, so this sequence fails the original strict convergence criterion.**

The physical norms are equal-point RMS values of $H = {}^{(3)}R + K^2 - K_{ij}K^{ij}$ and $|M|_\gamma = (\gamma_{ij}M^iM^j)^{1/2}$, with $M^i = D_jK^{ij} - D^iK$, in units M_0^{-2}, $M_0 = m_+ + m_- = 1$. They are not the internal sine-weighted conformal residual with endpoint rows. A supplemental combined exterior RMS $[\langle H^2 + |M|_\gamma^2\rangle]^{1/2}$ falls from 2.83×10^{-7} to 1.41×10^{-7} to 1.03×10^{-7}. This post-study summary does not replace the failed regional trend gate.

Necessary native solve times are 41.8, 52.7 and 140.3 seconds, after 60.6, 87.3 and 116.6 seconds of setup, with 8, 5 and 6 Newton steps. The first grid starts from zero; later grids reuse the preceding field as an initial guess. These are workflow records, not a matched-accuracy speedup measurement. No multigrid hierarchy is adopted: the retained stalled trumpet residual is predominantly grid-scale, and a coarse-error bottleneck has not been demonstrated. The bounded overintegrated, two-sweep modal and left-GMRES attempts also fail fine-grid linear targets; their optional implementations remain off by default.

The finest QI checkpoint passes separate-process producer/CPU-consumer sampling at 2454 points, with identical fields and metric gradients. The first AthenaK attempt stops before the finder because its Python proof validator recognizes only older basis identifiers. The compatibility fix explicitly admits C4 and axis-tau identifiers while retaining exact checkpoint, basis, map, image and field bindings; four regression tests pass, including rejection of unknown identifiers and invalid matching maps. The passed migration is reused, without a repeated sampling run.

AthenaK then passes harmonic orders $8, 12, 16$ and an independent quadrature increase from $n_\theta = 32$ to 48, with zero evolution. ADM/Z4c import error is 4.08×10^{-16}. The final measured properties are:

Hole	M_H	M_{irr}	χ_{coord}	Expansion RMS
+	0.615673713	0.604595771	0.370895278	2.04×10^{-8}
−	0.416086392	0.409618329	0.345813441	5.63×10^{-9}

All angular orders meet the 10^{-7} expansion bound. Across all consecutive angular and quadrature refinements, maximum mass-relative and dimensionless spin-vector changes are 2.00×10^{-12} and 2.78×10^{-11}. After the observed refinement buffer, the continuous modified-ball enclosure margins are 0.03205 and 0.01867. These are coordinate rotation-integral spins, not approximate-Killing-vector estimates, and angular stability does not measure the input solution's continuum error. Finder times are 47.5, 36.5, 38.4 and 43.8 seconds with 16 OpenMP geometry threads.

These import and horizon checks are diagnostic because the input sequence failed its strict bulk trend criterion. They neither qualify the trumpet family nor upgrade the QI checkpoint's diagnostic acceptance. Exact configurations, raw physical and horizon arrays, frozen-image hashes, figures and commands are retained in `validation/trumpet/regularity-family-control/`.

IV. ENDPOINT REGULARITY AND LIU CHART UPDATE

The current moderate study retains the wider QI formulation above. Host background construction now uses algebraically normalized seed metrics, `expm1` attenuation complements and compensated trace evaluation. An exact constant transverse Schwarzschild conformal metric exposes spurious old derivative magnitude 3.34×10^{-6} on Perlmutter (6.94×10^{-3} on the local compiler) at small radius; the stable expression gives exactly zero. The unchanged-map prefix still has bulk Hamiltonian RMS 6.47×10^{-9} and 8.42×10^{-9} on grids 160 and 192. It fails the strict trend, so the 224 setup is stopped and retained. This algebraic correction alone does not establish physical convergence. The original batch seed exports for QI and trumpet remain bitwise identical in 516 compared values. Background/cache/sampler arithmetic now uses stable equivalent expressions, so the historical sampler bitwise claim is not transferred to this numerical change.

The original 224-grid scalar radial tail is concentrated toward infinity; a degree-eight endpoint window reduces its retained high-degree band to 0.143 of the original. The vector tail is concentrated toward the inner boundary/transition; suppressing that end leaves 0.00128 of its band. These are localization measurements, not inferred logarithmic powers. A frozen 160-grid field sampled at twice the azimuthal resolution has interleaved physical momentum-component RMS below 3.52×10^{-12}, too small to explain the observed 10^{-7} bulk

momentum errors. No further azimuthal-resolution campaign is warranted.

A squared-infinity-distance map passes derivative and sampler checks but fails its first 160-grid linear solve (true relative residual about 0.368 at 512 iterations). No finer level is run. A dual-endpoint positive-density map instead preserves first-order compactification. With $s \in [0,1]$,

$$t(s) = \lambda s + bs^5 + cF(s), \qquad b = (2-\lambda)/5, \quad c = 1-\lambda-b,$$
$$F(s) = s^{15}[136(1-s)^2 + 17s(1-s) + s^2].$$

Here F is the Beta$(15,3)$ CDF. The selected $\lambda = 0.05$ gives $t'(0) = 0.05$, $t'(1) = 2$, versus 0.2 and 5 for the original map. Polar stretch is 3. The free data, inner window, equations and acceptance gates are fixed. Independent mapped-derivative, polynomial radial-inverse and Cartesian complex-step sampler checks pass; they do not imply binary acceptance. The ordered 16-thread physical sampler returns bitwise-identical fields to the serial sampler. No multigrid is adopted without a measured coarse-error bottleneck. All map options remain off by default.

The dual-endpoint 160-grid solve converges internally in 9 Newton and 51 Krylov iterations (46.0 seconds), but physical bulk momentum RMS is 2.14×10^{-5}, failing the original 10^{-6} bound. Refinement stops. The near-hole Hamiltonian RMS improves to 3.68×10^{-8}, but this does not compensate for a failed bulk gate. Frozen-field vector-mode localization places the error in low modes; the filtered fields are diagnostic only.

A separate candidate restores the original Mobius0.2/polar2 maps and imposes necessary Cartesian smoothness inside the flat plateau. For bounded Cartesian corrections, the constant-coefficient elliptic interior has removable puncture singularities. For $m \le 4$, at the two foci,

$$[\partial_t - 2\sigma\partial_\eta - (m+1)]P_m(0,\sigma) = 0, \qquad \sigma = \pm 1.$$

This follows by expanding $u_m/\rho^m \propto (1-t)^{m+1}P_m$ near a focus; it is distinct from the existing $m \ge 5$ axis rows. Exactly two collocation rows per low mode are replaced. No interior window or exterior equation changes. The normalized interpolation rows have unit L^1 coefficient norm. The preconditioner makes the same two-row replacement using

$$x = A_0^{-1}b + A_0^{-1}E(WA_0^{-1}E)^{-1}(b_{\text{boundary}} - WA_0^{-1}b).$$

Solving contexts require a positive flat interior plateau and zero far attenuation. Otherwise

the far-reference scalar correction can invalidate homogeneous rows on P alone. Sampling and raw operator contexts are exempt. CPU and GPU share the host factors and weights; compensated GPU moment reductions and rank-two corrections run in kernels. A prescribed smooth Cartesian field and a symbolic compatibility polynomial give normalized defects below 9.24×10^{-17} at 96/144 points. An independent dense inverse agrees to 3.65×10^{-15}. A small coupled binary converges in 9 Newton/32 Krylov steps. These controls establish the new boundary algebra, not three-grid physical convergence.

The first-compatibility (C1) sequence meets every absolute bound but fails both strict bulk trends. Its independent higher-accuracy observer gives:

n_A	Near H	Near M	Bulk H	Bulk M
160	3.13×10^{-7}	8.50×10^{-8}	4.38×10^{-10}	3.89×10^{-8}
192	1.45×10^{-7}	4.10×10^{-8}	1.20×10^{-9}	5.02×10^{-9}
224	4.89×10^{-8}	1.83×10^{-8}	2.24×10^{-9}	7.77×10^{-9}

The finest bulk momentum error improves by 21.9 over the original sequence, but that is not a passed convergence study. Metric and curvature Cauchy changes decrease, with finest scaled maxima 4.79×10^{-11} and 2.95×10^{-9}, respectively; charges change by less than 4.08×10^{-10}. These cannot replace the failed constraint criterion.

This observer uses sixth-order Cartesian differences at four times the original fixed step, on the same 66 exterior points. It was declared after inspecting the coarse C1 result, before inspecting finer results; no original FD4 verdict is replaced. Every saved field first reproduces the original FD4 observer. No new elliptic solves are needed. Rational-stencil moment, manufactured-curvature and exact Brill–Lindquist controls pass; the latter has near/bulk Hamiltonian RMS $7.10 \times 10^{-11}/8.02 \times 10^{-11}$. The new observer therefore confirms that finite-difference error alone does not explain the HiSpID bulk trend.

The remaining second-order Cartesian condition follows from $Q_{tt} = 4Q_{\eta\eta} + 4\sigma Q_\eta$ for $Q = (1-t)^{m+1}P_m$ at each focus:

$$[\partial_t^2 - 4\partial_\eta^2 - 2(m+1)\partial_t - 4\sigma\partial_\eta + m(m+1)]P_m(0,\sigma) = 0.$$

Retained C1 solutions have normalized C2 defects 0.00276,0.00379,0.00322, whereas their C1 defects are below 1.67×10^{-15}. The second-order candidate replaces four rows total and uses the same exact rank-four boundary inverse on CPU and GPU. A smooth Cartesian

field and compatible symbolic polynomial give defects below 2.11×10^{-16}; its independent dense inverse agrees to 3.24×10^{-15}. The original compactification, physical free data and acceptance gates are fixed. These are necessary smoothness identities, not an imposed analytic asymptotic expansion at compactified infinity. Results of the new physical sequence are recorded separately below.

A. Finest retained C1 checkpoint in AthenaK

The 224/448/16 checkpoint passes separate-process GPU-producer/CPU-consumer migration: every sampled field and metric derivative is bitwise identical at 2454 off-grid and trial-horizon points. AthenaK imports the actual checkpoint and measures two distinct horizons at angular orders 8,12,16, followed by an independent 32-to48-point polar quadrature check at order 16. No evolution is performed; the finder samples the checkpoint's native continuous geometry.

Hole	M_H	M_{irr}	Coordinate χ	Expansion RMS
1	0.615673713	0.604595771	0.370895278	2.041×10^{-8}
2	0.416086392	0.409618329	0.345813441	5.626×10^{-9}

Maximum mass and dimensionless spin-vector changes across the refinements are below 3.50×10^{-13} and 4.04×10^{-12}. The verified modified-interior enclosure margins, after the observed shape-refinement buffer, are 0.03204669 and 0.01867393. These positive margins and horizon property measurements do not override the failed physical bulk trend. Coordinate angular-momentum integrals are not a binary gauge-invariant or approximate-Killing-vector spin validation. Full grid evolution is untested.

B. Second compatibility is not selected

The added condition is algebraically verified but fails its physical trial:

n_A	Bulk H (FD4)	Bulk M (FD4)	Newton/Krylov
160	1.96×10^{-9}	1.05×10^{-6}	5/35
192	6.30×10^{-9}	1.78×10^{-6}	5/33

The original absolute bound is unchanged. After the coarse failure, a single predeclared refinement test required a clear improvement before continuing any 192/224/256 sequence. It fails; no 224 or 256 C2 trial is run. The first-order 160/192/224 study remains the primary record. A successful internal solve and necessary smoothness identities do not establish a useful global tau discretization. Regularity/asymptotic representation remains an unresolved numerical issue, rather than a proven single cause of the bulk error.

`HISPID_FOCUS_TAU` remains off by default. When enabled, `HISPID_FOCUS_ORDER=1` selects the better tested v9 system; order 2 selects the experimental v10 system. They share the operator and boundary-inverse code. Current order-one weights differ from the frozen C1 implementation by at most 3.47×10^{-18}; direct boundary residuals on all three retained fields stay below 1.60×10^{-15}. No new physical acceptance is inferred from this compatibility control.

C. Matched-count Bowen–York Hamiltonian control

The BY control uses the unchanged vanilla constraint equation with the previously baseline-validated modal CUDA backend. Counts, bare masses, separation, isolated Lorentz-transformed input spins, and $P = m\Gamma v$ are matched. Physical free metrics and binary horizon properties differ. Both cases use exactly the same stored Cartesian observer points and fourth-order finite-difference steps, with the physical definition $H = R + K^2 - K_{ij}K^{ij}$. These are equal-point RMS norms in the same M_0^{-2} units. Native collocation coordinates and internal stopping weights differ.

n_A	BY near H	BY bulk H	Reference $\|F\|_\infty$	Newton cap
160	9.71×10^{-10}	1.18×10^{-9}	1.03×10^{-10}	24
192	1.08×10^{-9}	1.46×10^{-9}	1.52×10^{-10}	24
224	1.19×10^{-9}	1.54×10^{-9}	1.94×10^{-10}	8

Here $n_B = 2n_A$, $n_\phi = 16$ and the requested native tolerance is 10^{-12}. All original-reference stopping gates fail at a roundoff plateau. The finest control is capped at 8 Newton steps after the earlier controls show repeated floor-level updates. The physical comparison is diagnostic; it is not an iteration-count or matched-accuracy performance measurement. The

nearly constant physical H values help characterize the common verifier scale, but are not alone proof of the exact finite-difference error floor.

The same saved BY solutions, reconstructed without any Newton iteration, give sixth-order physical Hamiltonian RMS:

n_A	BY near H (FD6)	BY bulk H (FD6)
160	8.03×10^{-11}	9.94×10^{-11}
192	8.67×10^{-11}	1.03×10^{-10}
224	8.09×10^{-11}	1.11×10^{-10}

The independent exact conformally flat calibration has a comparable floor. Thus the BY physical Hamiltonian comparison is already verifier limited at these counts, whereas the near-hole HiSpID errors remain much larger. Neither raw collocation residuals nor different physical free data are claimed equivalent merely because the observer normalization is shared.

Large matched counts initially hit a guard before Newton. The capacity extension retains checked size arithmetic, aggregate/device budgets and the legacy allocation API's 4GiB ceiling; an explicit execution budget permits a larger factor bank. A supported BY reference solution and all 33 recorded arrays/derivative exports remain bitwise identical after this change. The equations, residual and linearization are unchanged. Raw observers, failed stopping records and the preservation control are retained with the HiSpID evidence. No multigrid change is justified by these few-iteration linear solves; repeated host factor construction remains a separate cost.

D. Liu–Etienne–Shapiro coordinates

The previous code had QI and $R_0 = M$ trumpet seeds. The newly selectable Liu family implements Eqs. 12 and 14 of Liu, Etienne and Shapiro, Phys. Rev. D80,121503 (2009):

$$r_{\rm BL} = r\left(1 + \frac{r_+}{4r}\right)^2, \quad r_+ = m + \sqrt{m^2 - a^2}, \quad r_H = \frac{r_+}{4} \longrightarrow \frac{m}{4}.$$

Ruchlin et al. also discuss this chart as LES coordinates in their Eq. 16; their construction permits QI, LES and a separate fisheye chart. This is a wormhole chart, not the existing trumpet. The finite formal extremal-radius limit does not imply support for exact $|\chi| = 1$,

which remains rejected. Spin-axis rotations and full spacetime Lorentz boosts are shared with QI; radial tensor pullbacks and throat-limit curvature are handled analytically. Independent pullbacks agree to 6.18×10^{-15} through the throat and both bridge sides. Vacuum checks cover $\chi = 0, 0.39, 0.99$, unboosted and with a generic moderate boost.

An exact isolated unboosted $m = 1$, $\chi = 0.99$ checkpoint passes separate-process producer/CPU-consumer migration and AthenaK horizons at orders 8,12,16, with different initial-radius guesses. Maximum mass and spin errors are 5.87×10^{-13} and 1.37×10^{-12}; expansion RMS is below 9.84×10^{-8}. Finest coordinate radius is 0.285266965 versus analytic0.285266840. Stationary area and spin integrals are more accurate than surface location. No binary or extreme-boost Liu qualification follows from this exact-seed control. The moderate binary study continues to use QI. A separate moderate Liu binary pilot at 160/320/16 converges internally in 8 Newton/29 Krylov steps. Near Hamiltonian/momentum RMS are $3.47 \times 10^{-7}/5.48 \times 10^{-8}$; bulk values are $9.67 \times 10^{-10}/4.33 \times 10^{-8}$. Absolute exterior bounds pass, but one resolution establishes neither convergence nor horizon enclosure. The fixed three-grid formulation is not switched to this family. Evidence and commands are in `validation/trumpet/infinity-regularity/`.

V. RETAINED TRUMPET BINARY EVIDENCE

The retained pre-tau entries below are independent physical RMS errors, in the declared normalization. The moderate case has masses 0.6/0.4, separation 6, and generic spins and boosts; exact parameters are in `examples/trumpet_configs.py`. The spin target has equal masses 0.5, aligned seed $\chi = 0.99$, and separation 12. The head-on target has opposite seed $\Gamma = 10$, masses 0.5, and separation 25. The latter has not yet been calibrated with qualified binary irreducible masses.

Case / grid	Near H	Near M	Bulk H	Bulk M
Moderate $192\times384\times16$	1.13×10^{-6}	3.77×10^{-7}	5.76×10^{-8}	4.42×10^{-7}
Moderate $224\times448\times16$	2.08×10^{-7}	1.65×10^{-7}	8.71×10^{-8}	4.44×10^{-7}
Moderate $256\times512\times16$	1.70×10^{-7}	1.39×10^{-7}	2.62×10^{-7}	2.86×10^{-7}
Spin $160\times320\times32$	3.15×10^{-6}	1.44×10^{-5}	8.30×10^{-5}	3.00×10^{-4}
Boost $256\times320\times8$	8.23×10^{-4}	2.42×10^{-2}	4.46×10^{-6}	1.27×10^{-4}
Boost $512\times320\times8$	1.02×10^{-4}	6.12×10^{-3}	1.07×10^{-6}	3.75×10^{-5}

The retained C2 moderate sequence fails because bulk errors do not decrease. A diagnostic removal of the scalar $m = 6$ pair reduces finest-grid bulk Hamiltonian RMS from 2.62×10^{-7} to 5.20×10^{-9}; the mode also has a forbidden ρ^4 axis term whose coefficient grows with resolution. This diagnoses sensitivity, not permission to filter a physical solution. Tighter stopping and smaller observer steps do not remove the defect.

A. Regularity correction and selected solver

The current opt-in construction retains C2 modal unknowns and imposes $P(0,\eta) = P(t,\pm1) = 0$ for $m \geq 5$ through independent endpoint rows. It avoids dividing sources by additional vanishing axis factors. A scalar manufactured case converges to 2.40×10^{-14}; under an absolute 10^{-16} source perturbation, its finest-grid field noise is 2.17×10^{-11} versus 3.43×10^{-6} with factored C4 unknowns. The forcing has an independent Cartesian-Hessian check. Native CPU/CUDA endpoint-row reconstruction, directional JVP and checkpoint controls pass; the largest CUDA row and JVP errors are 3.33×10^{-16} and 1.00×10^{-11}, respectively.

The preconditioner must enforce the same endpoint rows. Exact radial rows fit its existing dense blocks. Exact polar rows use a rank-$2n_A$ Woodbury update, with batched response construction and shared CPU/Kokkos application. An independent dense-matrix test including pivots and nonzero boundary data gives solution error below 6.7×10^{-15}. Small coupled solutions agree between CPU and CUDA to 2.82×10^{-13}; residual/JVP witnesses are unchanged by the preconditioner modification. Allocation guards include the extra matrices.

The full $192\times384\times16$ moderate solve in job 59420346 now converges in 5 Newton/53

Krylov steps, taking 65.4 seconds after 65.5 seconds of setup. All Newton steps are undamped. The independent near-hole H/M RMS errors are $1.128 \times 10^{-6}/3.789 \times 10^{-7}$; bulk errors are $5.739 \times 10^{-8}/4.548 \times 10^{-7}$. Only the near-hole Hamiltonian RMS exceeds its bound. The workflow takes 246.6 seconds and peaks at 6.025 GiB host RSS. This grid remains diagnostic. The $224 \times 448 \times 16$ continuation initially fails a late linear solve, ending at relative residual 0.0245 after 2400 iterations against the requested 0.001. Resuming the saved iterate with standard forcing 0.1 takes one additional Newton step and 119 Krylov iterations (36.8 seconds), reaching nonlinear maximum 3.35×10^{-13} against the unchanged 10^{-12} tolerance. Independent near H/M RMS errors are $2.071 \times 10^{-7}/1.651 \times 10^{-7}$; bulk errors are $5.023 \times 10^{-8}/4.336 \times 10^{-7}$. All per-grid physical bounds pass. The subsequent 240×480 run with restart 24 fails a late linear solve (relative residual 0.873 after 2400 iterations); bulk momentum RMS is 1.72×10^{-6} and the nonlinear maximum is 1.62×10^{-9}. Resuming that saved state with restart 64 also fails after 2400 iterations: the true relative residual improves to 0.121, but exceeds the required 0.1. No Newton update is accepted. This retry takes 430.7 solve seconds and 692.8 workflow seconds; peak host RSS is 11.54 GiB. The allocation is released and the 256×512 level is not launched. Further work must diagnose the linear operator/preconditioner rather than relaxing the final criteria. The passing 224-grid result is retained. All levels use $n_\phi = 16$, the same native image, and unchanged nonlinear and physical acceptance criteria. Each completed grid must pass nonlinear and physical bounds before continuing, and the final assessment checks decreasing errors and charges. A matching CPU sampler and AthenaK consumer built successfully in job 59420409; new-checkpoint import and horizon measurements remain to be run. The option and distinct checkpoint representation remain experimental.

A bounded 64-step replay diagnoses genuine slow linear convergence: the true relative residual is 0.922, and recurrence versus explicit residual differ by only 1.97×10^{-13} relatively. A manufactured matrix witness at the failing resolution is recovered by all host modal factors to 1.12×10^{-9} or better. This checks host factor accuracy for that witness, not the approximation quality or the GPU inverse application. The subsequent actual-RHS comparison finds a 7.48% global GPU/host preconditioner discrepancy, dominated by mode-4 momentum components. Before this correction, residual energy is 99.72% momentum and 70.92% modes 0–1; tau rows contribute negligibly. Feeding identical projected input to both inverse routines reduces the discrepancy to 2.98×10^{-16}, isolating Fourier projection before

near-axis row scaling. An opt-in compensated projection preserves subtraction and product rounding errors during summation. Exact dyadic cancellation controls pass on the host; the small nonlinear control retains five Newton/53 Krylov iterations and changes sampled fields by only 3.12×10^{-13} (scaled), with bitwise unchanged residual/JVP witnesses. GPU cancellation controls also pass, and the corrected actual-state GPU/host inverse difference is 2.82×10^{-16}. A full Newton step now converges in 319 Krylov iterations, but the next inner solve stops at relative residual 0.306 after 2400 iterations. The nonlinear maximum improves to 2.17×10^{-11} but still fails 10^{-12}; bulk momentum RMS 2.53×10^{-6} also fails. A continuation with inexact-Newton forcing $\eta = 0.5$ accepted one step but the next linear solve stalled at relative residual 0.547 after 2400 iterations. Bulk momentum RMS remained 2.53×10^{-6}; this continuation also fails acceptance. For $Js = -r + e$ and $\|e\| \leq \eta\|r\|$, $r^T Js \leq -(1-\eta)\|r\|^2$; the existing line search verifies actual decrease. An existing BiCGStab-backend probe of this frozen system diverges to true relative residual 1.73×10^{103} after 1200 iterations (2402 JVPs, 401.8 seconds), with no accepted Newton update. Optional augmented GMRES retains three previous-cycle correction directions; CPU and CUDA manufactured controls pass, and a small nonlinear solve agrees between backends to 2.57×10^{-13} in sampled fields. Its additional workspace raises the 240-grid allocation bound to 61.27 GiB, within 64 GiB. On the frozen failing system it reaches relative residual 0.471 after 2400 iterations (421.4 seconds), still above 0.1, with no Newton update. Thus neither alternative resolves the large-system stagnation. After the projection correction, the failed GMRES residual instead has 98.53% momentum energy spread over modes 0–7; modes 0–1 carry only 26.57%. A separate evaluation of the retained 240-grid nonlinear state finds interior residual concentration: the outermost eight radial and polar layers carry only 4.99×10^{-6} and 3.29×10^{-8} of total squared residual, respectively. These observations do not support an endpoint-only repair or another Krylov-method sweep. The same-checkpoint CPU/CUDA residual comparison gives $\|F_{\mathrm{CPU}} - F_{\mathrm{CUDA}}\|_2 = 2.11 \times 10^{-13}$, only 1.42×10^{-4} of the CUDA residual norm 1.48×10^{-9}. Backend-specific evaluation differences therefore do not account for the dominant stalled residual; common discretization or precision errors remain possible. Neither backend is assumed to be exact. Reanalysis of these saved arrays finds 93.60% of weighted residual energy in the last eight radial DCT modes and 78.56% in the last eight polar modes; median degrees are 238 and 477 on 240 and 480 nodes. Both backends agree. This identifies grid-scale stagnation and motivates improving the spectral operator's low-

order preconditioner, without establishing the cause or justifying residual filtering. The opt-in spectral radial preconditioner now uses exact polynomial radial differentiation in the existing dense blocks, retaining polar finite differences and endpoint conditions. The residual and Jacobian-vector product are unchanged; the default backend and Bowen–York path are unaffected. Compact CPU assembly avoids dense entries in per-point sparse containers. Added host derivative/map tables require $16n_r^2 + 24n_r$ bytes, included in the aggregate memory guard. Manufactured radial controls give worst scaled errors 7.00×10^{-13} on CPU and 3.66×10^{-14} on CUDA. The small binary requires five Newton steps and 48 Krylov iterations (previously 53), with CPU/CUDA sampled-field difference 2.39×10^{-13}. These controls qualify the implementation, not fine-grid effectiveness.

The final frozen-state GPU probe (Perlmutter job 59426128) uses the same 240-grid checkpoint, GMRES restart 64, relative target 0.1, and budget 2400. It fails with true relative residual 0.78667352, 2439 Jacobian-vector products, and 448.51 seconds of diagnostic solve time. No accepted Newton update or improved physical solution follows. The option remains disabled by default; it is not recommended as a demonstrated remedy for the fine-grid stall. Evidence is retained in `validation/trumpet/c4-experiment/axis-tau/polar-exact/spectral-radial/gpu-v1/probe.json`.

Final nonlinear and physical criteria remain unchanged. Binary acceptance remains open.

Retained alternative	Decisive outcome
Factored C4 unknowns	Manufactured controls pass; fine binary fails line search. Tighter forcing, longer backtracking and coarse continuation do not cure it.
Preconditioner scaling	Two prototypes fail the small coupled control; rejected.
Stable scalar-source assembly	Reduces measured angular source roundoff, but fine replay still needs damping 1/2048 with negligible residual decrease.
Monotone FD preconditioner	Small solve improves; fine replay still needs 1/1024 damping with negligible residual decrease.
Local polar endpoint rows	Fine linear solve fails after 2400 iterations; exact radial rows alone also fail (relative residual 0.0162).
Exact radial and polar rows	Fine first linear solve passes in 6 iterations; full nonlinear solve converges; coarse physical error still requires refinement.

These experiments are not a new backend benchmark campaign. Full configurations, failed states and precision/transfer diagnostics remain under `validation / trumpet / c4-experiment/`; none are discarded or promoted to physical acceptance. The original C2 default and successful historical data remain available.

The retained boost map is $(\lambda, \kappa) = (0.03, 3.5)$. Radial refinement from 160 to 256 improved its errors by factors 2–6. Stronger focusing, $\lambda = 0.003$, worsened both independent errors and radial tails and is rejected. Radial-512 mapped-operator controls pass at normalized error 5.49×10^{-10}. After the original timeout, the reduced-factor retry completed: relative to radial 256, near-hole Hamiltonian/momentum RMS errors improve by factors 8.09/3.95 and bulk errors by 4.18/3.39. Extrapolated energy changes by 5.48×10^{-6} relatively. These improvements do not meet the physical bounds; the diagnostic horizon check also fails.

VI. HORIZON MEASUREMENTS AND COMPUTATIONAL COST

The earlier pre-tau trumpet moderate 256-grid checkpoint passes import and horizon angular/quadrature checks, but remains diagnostic because of its constraint sequence. Its component irreducible masses are 0.5887780114 and 0.3935773863.

The spin-target diagnostic checkpoint passes orders $12, 16, 20$ and an independent quadrature increase. Each component has $M_H = 0.500017851900$, $M_{\mathrm{irr}} = 0.377763585476$, and coordinate $\chi = 0.989929520988$, with expansion RMS 5.98×10^{-9}. Both retained surfaces enclose the modified regions. These are coordinate rotation-integral spins, not approximate-Killing-vector estimates. Reusing passed harmonic shapes strictly as initial guesses reduces order-12/16 runs from 671/1227 seconds to 35.5/50.4 seconds; all geometry and acceptance checks are recomputed.

For boosted horizons an optional spatial chart $x = c + J(y - c)$, $J = I - vv^T/(1+\sqrt{1-v^2})$, makes the exact isolated seed surface spherical. Tensors and derivatives are pulled back consistently; physical rotation generators become $J^{-1}E_iJ$. This changes no slice. The isolated $\Gamma = 10$ control passes orders $4, 8, 12$: relative area error $\leq 7.22 \times 10^{-11}$, shape error $\leq 1.13 \times 10^{-10}$, and expansion RMS $\leq 4.78 \times 10^{-11}$. The solved binary instead stalls near 1.8×10^{-5} at both orders 8 and 16; it has no qualified horizon result. At radial 512 the order-8 expansion floor falls to 9.49×10^{-6} after 1000 iterations, still above 10^{-7}. Import

succeeds with ADM/Z4c relative error 4.39×10^{-16}. The driver stops at the failed surface; its area does not supply a qualified irreducible mass for separation calibration.

Necessary production solves, not a matched-accuracy benchmark, took 94.2 seconds (315 seconds total, 4.87 GiB peak process RSS) for the listed spin case and 161.3 seconds (273 seconds total, 3.53 GiB RSS) for the listed boost case. Host RSS excludes GPU allocation. The radial-512 run retains the existing 64 GiB aggregate guard; its conservative estimate is 63.74 GiB. The completed radial-512 retry took 3119.6 seconds in the nonlinear solve (3294.6 seconds total, 5.59 GiB peak host RSS), with 10 Newton steps and 352 Krylov iterations. It used the original column-wise transfer construction.

The longer radial-512 attempt exposed a host factor-construction bottleneck: a live stack snapshot found active threads in the modal preconditioner's column-wise triangular solves. The qualified opt-in implementation constructs $A^{-1}\,\mathrm{diag}(U)$ with two matrix-right-hand-side triangular solves, retaining the same LU factors, pivots and block recurrence. On a 512-row pivoted control, the original-matrix residual is 7.41×10^{-16} and the old/new transfer difference is 5.11×10^{-16}. Necessary kernel timings on Perlmutter are 0.556 versus 0.107 seconds (a factor of 5.19); **this is not a whole-solver speedup measurement**. A local small-solve field/gradient comparison agrees within 1.31×10^{-16}, and the separate CUDA axisymmetric control passes. The option remains off by default. A continuation command pins the tested image and preserves the radial-512 configuration. It was not launched because the original run completed; future necessary solves can use the qualified optimization. Evidence is in `validation/trumpet/batched-transfer/` and `validation/trumpet/axial-factors/stack-snapshot.txt`.

VII. REPRODUCIBILITY AND REMAINING WORK

The native and consumer branches are `codex / hispid-trumpet` and `codex / hispid-trumpet-pgen`; neither is merged. Detailed evidence, failed attempts, hashes and executable identities are under `validation / trumpet/`, with the audit trail in `docs/TRUMPET_IMPLEMENTATION.md`. Key records are `convergence/assessment.json`, `spin-focused-map / horizon160-warm/`, `affine-horizon / exact / horizons-v4/`, and `axisymmetric/radial-comparison.json`. Commands are retained beside each run. The original performance report is preserved without extending its campaign.

Completion still requires an accepted moderate physical-convergence sequence, resolution of the extreme binary constraint errors, qualified boost-binary horizon measurements and measured-$M_{\rm irr}$ separation calibration, followed by a final evidence audit. Pending runs and failed gates are not counted as validation. This report is an interim snapshot, not completion of the goal.

References: Dennison–Baumgarte–Montero (2014); Ruchlin et al., high-spin/high-boost puncture data; boosted trumpet construction reference.